



Detection of slowly varying spoofing using weighted Kalman gain in GNSS/INS tightly coupled systems

Xiaoqin Jin¹ · Xiaoyu Zhang¹ · Shoupeng Li¹ · Shuaiyong Zheng²

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Abstract

Global Navigation Satellite System (GNSS) is vulnerable to intentional spoofing attacks, particularly those that vary slowly over time. These attacks aim to deceive receivers by introducing subtle changes to the received signals, making them hard to detect using traditional methods. To tackle this challenge, we propose an improved slowly varying spoofing detector that uses a weighted Kalman gain to enhance the sensitivity of the extended Kalman filter (EKF) to slowly varying spoofing. Our detector addresses the limitations of the conventional EKF, which does not account for the impact of spoofing on the innovation offsets, leading to rapid induction of the filtering results. By weighting the EKF gain based on the normalized distance between the test statistic of each satellite and the detection threshold, our proposed detector mitigates harmful satellite innovations and accumulates the innovation offsets caused by spoofing. Simulation and experimental results demonstrate that the proposed detector achieves a higher detection probability and sensitivity compared to existing methods. Our proposed detector offers a novel approach to detect slowly varying spoofing and represents a significant contribution to the field of GNSS security.

Keywords GNSS · Slowly varying spoofing detection · Tightly coupled navigation · Weighted Kalman gain · Detection sensitivity

Abbreviations

AVP	Attitude, velocity, and position
DOF	Degree of freedom
ECEF	Earth-centered, earth-fixed
EKF	Extended Kalman filter
FAR	False alarm rate
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
INR	Interference-to-noise ratio
INS	Inertial navigation sensor
PDF	Probability density function
PPS	Pulse per second
ROC	Receiver operating characteristic
TCS	Tightly couple system

Introduction

Global Navigation Satellite System (GNSS) is widely used in a range of applications, including transportation (e.g., navigation systems in cars, trains, and planes), location-based services (e.g., ride-sharing apps, mapping applications), and communication systems (e.g., satellite phones, emergency communication devices) (Jin et al. 2022). However, GNSS signals are vulnerable to intentional spoofing attacks, where false signals are broadcast to deceive GNSS receivers (Psiaki and Humphreys 2016). Slowly varying spoofing signals can gradually mislead receivers into providing incorrect position, velocity, and timing information, making it essential to develop high-sensitivity spoofing detectors (Gao and Li 2022).

Slowly varying spoofing detectors can be broadly classified into two categories: receiver autonomous detectors and detectors based on other sensors (Meng et al. 2022; Qiao et al. 2023). Receiver autonomous detectors employ spoofing detection techniques such as signal quality monitoring, which includes evaluating cross-correlation function distortion and detecting multiple correlation peaks (Wang et al. 2022; Shang et al. 2023a, b). They also utilize signal

✉ Xiaoyu Zhang
zhangxiaoyu@nankai.edu.cn

¹ College of Artificial Intelligence, Nankai University, Tianjin 300350, China

² School of Integrated Circuit Science and Engineering, Tianjin University of Technology, Tianjin 300384, China

parameter monitoring, which involves assessing parameters such as signal strength, Doppler, and carrier-to-noise ratio (Gross et al. 2019; Chen et al. 2022). Additionally, these detectors use multi-receiver or multi-antenna observations (Abdelaziz et al. 2019; Seo et al. 2018) and implement receiver autonomous integrity monitoring (Singh et al. 2022). On the other hand, detectors based on other sensors rely on measurements from non-GNSS sensors, including inertial navigation sensors (INS), odometers, geomagnetic sensors, high-precision atomic clocks, barometers, light detection and ranging sensors, and so on (Gu et al. 2022; Vagle et al. 2018; Chang et al. 2022; Wei et al. 2022). INS-based detectors are commonly adopted due to their immunity to radio frequency interference and high reliability (Geng et al. 2023).

The INS-based detectors combine GNSS and INS using an extended Kalman filter (EKF) to generate innovations for constructing test statistics for spoofing detection (Shang et al. 2023a, b). There are three modes of integration: loosely coupled, tightly coupled, and deeply coupled (Boguspayev et al. 2023; Nouredin et al. 2009). Tightly coupled integration is widely adopted due to its higher accuracy compared to loosely coupled integration and its simpler implementation compared to deeply coupled integration. In a tightly coupled system, the innovations are determined as the discrepancies between the pseudoranges or pseudorange rates obtained from standalone GNSS and the prior estimates derived from the EKF (Tanil et al. 2018). In the absence of spoofing, the innovations are Gaussian variables with zero mean (Joerger and Pervan 2013). When spoofing first occurs, the pseudoranges or pseudorange rates from GNSS are contaminated, while the prior estimates from the EKF have not been entirely spoofed, and the differences between them are reflected in the innovation offsets, which can be detected (Zhang et al. 2022). Khanafseh et al. (2014) further assessed the performance of this spoofing detection method and confirmed its effectiveness. Tanil et al. (2017) expanded its application to aviation and validated its compliance with integrity requirements.

The spoofing-induced innovation offsets are typically very small for slowly varying spoofing, making it difficult for the test statistic constructed from innovations to exceed the detection threshold, resulting in undetectable spoofing (Liu et al. 2017). To address this issue, windowed detectors are employed, which accumulate the spoofing-induced innovation offsets over a time window to enhance the test statistic and enable detection (Tanil 2016). There are two types of windowed detectors: the windowed statistic detector, which accumulates the test statistics within the time window, and the windowed innovation detector, which accumulates the innovations (Zhang et al. 2022). Both types of detectors can accumulate the innovation offsets caused by spoofing. Currently, most of the research

focuses on the windowed statistic detector, with relatively less attention given to the windowed innovation detector. While Liu et al. (2019) verified through simulations that the windowed innovation detector exhibits higher detection sensitivity than the windowed statistic detector, they did not provide theoretical proof. We provide an analysis based on the interference-to-noise ratio (INR), which indicates the sensitivity of the different detectors. Moreover, we propose an enhancement to the windowed innovation detector by incorporating a weighted Kalman gain. The main contributions of our study are clarified as follows:

- (1) The higher detection sensitivity of the windowed innovation detector than the windowed statistic detector is derived from the perspective of improving the INR.
- (2) By analyzing the impact mechanisms of different spoofing cases on spoofing detectors, an accurate numerical definition of slowly varying spoofing is provided for the first time.
- (3) We propose an improved slowly varying spoofing detector using weighted Kalman gain. The conventional EKF solely relies on the model covariance to estimate the filter gain, neglecting the impact of spoofing on the innovation offset. As a consequence, the filtering results are susceptible to rapid induction by spoofing. To address this issue, our proposed detector weights the EKF gain by the normalized distance between the test statistic of each satellite and the detection threshold. This weighting approach ensures that observations from the satellites that have not been contaminated by spoofing are maximally included in the filter update, while mitigating the harmful satellite innovations that contaminate the filter and further accumulating the innovation offsets caused by spoofing to enhance the sensitivity of the detector.

Simulations and experiments at Nankai University were conducted to evaluate the efficacy of the proposed detector's spoofing detection performance. The results demonstrate that the proposed detector exhibits greater sensitivity to slowly varying spoofing attacks, thereby providing improved protection against such threats.

The rest of the paper is organized as follows. First, the filter model and basic spoofing detectors are reviewed. Then, the effect of the innovation offsets on the detectors is analyzed and an accurate definition of slowly varying spoofing is given. Next, the proposed spoofing detector is illustrated. Additionally, the simulations and experiments are provided, respectively, demonstrating the sensitivity advantages of the proposed detector. Finally, the conclusions and discussions are drawn.

Filter model and basic spoofing detectors

In this section, we first briefly review the EKF innovation-based spoofing detector. Then, two typical windowed detectors for slowly varying spoofing are described in detail, and their performance comparisons are derived.

Review of the extended Kalman filter innovation-based spoofing detector

In a typical Kalman filter, the common state equation and measurement equation are described as (Liu et al. 2023)

$$\mathbf{X}_k = \Phi_{k/k-1} \mathbf{X}_{k-1} + \mathbf{W}_{k-1} \quad (1)$$

$$\mathbf{Z}_k = \mathbf{H}_k \mathbf{X}_k + \mathbf{V}_k \quad (2)$$

where $\mathbf{X}_k$ and $\mathbf{X}_{k-1}$ are the state vectors at time k and $k-1$, respectively. $\Phi_{k/k-1}$ is the state transition matrix from time $k-1$ to k . $\mathbf{W}_{k-1}$ is the system noise, of which the covariance matrix is $\mathbf{Q}_{k-1}$. $\mathbf{Z}_k$ and $\mathbf{H}_k$ represent the measurement vector and matrix at time k . $\mathbf{V}_k$ is the measurement noise with the covariance matrix $\mathbf{R}_k$.

In a GNSS/INS tightly coupled system, the state vector contains the attitude error, velocity error, position error, gyro bias, accelerometer bias, clock bias, and frequency bias. A typical EKF innovation-based spoofing detector is shown in Fig. 1, including time update, spoofing detection, and measurement update. In the stage of time update, the state transition matrix $\Phi_{k/k-1}$ is computed based on the output of the INS (Groves 2008). This matrix is then used to obtain a priori state estimate $\hat{\mathbf{X}}_{k/k-1}$ and its corresponding covariance matrix $\mathbf{P}_{k/k-1}$. If GNSS measurements, specifically pseudorange and pseudorange rate, are available at time k , the predicted pseudorange and pseudorange rate can be calculated using the INS-derived position and velocity, as well as the observed positions and velocities of the GNSS satellites obtained from ephemeris data. The difference between the GNSS measurements and the predicted pseudorange and pseudorange rate represents the measurement vector $\mathbf{Z}_k$ at time k . Additionally, the measurement matrix $\mathbf{H}_k$ is computed based on the INS-predicted receiver position and

velocity, as well as the observed positions and velocities of the GNSS satellites. Furthermore, the innovation vector γ_k can be computed using the above information and is utilized for spoofing detection. If the spoofing attack is detected, the GNSS measurements cannot be trusted and are removed from the filter. Otherwise, the third stage measurement update is carried out based on the innovation vector, resulting in the gain matrix $\mathbf{K}_k$, a posteriori state vector $\hat{\mathbf{X}}_k$, and state estimate error covariance matrix $\mathbf{P}_k$.

As for the innovation vector, it can be used to detect spoofing due to its Gaussian distribution feature. The covariance matrix of γ_k is (Tanil 2016)

$$\mathbf{D}(\gamma_k) = \mathbf{A}_k = \mathbf{H}_k \mathbf{P}_{k/k-1} \mathbf{H}_k^T + \mathbf{R}_k \quad (3)$$

Conventional spoofing detection is based on a binary hypothesis: spoofing-absent H_0 and spoofing-present H_1 . In these two conditions, the mean vector of γ_k meet

$$\begin{aligned} H_0 : E(\gamma_k) &= 0 \\ H_1 : E(\gamma_k) &= \mu \end{aligned} \quad (4)$$

where μ is the innovation offset caused by spoofing, which can also be regarded as a kind of energy of the spoofing. Based on such a hypothesis, a spoofing detection function is defined as (Tanil 2016)

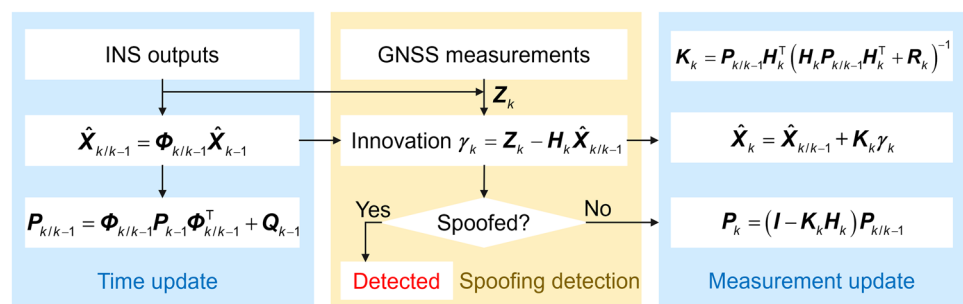
$$\lambda_k = \gamma_k^T \mathbf{A}_k^{-1} \gamma_k \quad (5)$$

where the length of γ_k is M , and the test statistic λ_k is a Chi-squared variable with the degree of freedom (DOF) of M . According to the Neyman–Pearson criterion, the relationship between the false alarm rate (FAR) P_F and the detection threshold T_D is

$$P_F = \int_{T_D}^{\infty} \chi^2(\lambda, M) d\lambda \quad (6)$$

where λ is the samples of λ_k . During the detection process, considering λ_k as a test statistic, when λ_k exceeds T_D , a spoofing is detected, and an alarm is given. The aforementioned spoofing detector is a kind of snapshot detection method that only utilizes the innovation at one epoch and

Fig. 1 Three stages of an EKF filter and spoofing detector. The three subplots correspond to the time update (left), spoofing detection (middle), and measurement update (right) stages



applies to suddenly varying spoofing. Moreover, to realize a higher detection sensitivity for slowly varying spoofing, windowed detectors, which are effective in the accumulation of spoofing energy, have been widely applied.

Windowed spoofing detectors and performance comparison

There are two main windowed detectors: the windowed statistic and innovation detectors. Here, the length of a sliding time window is set to N . The windowed statistic detector is represented as (Tanil et al. 2016)

$$\lambda_k^W = \sum_{j=k-N+1}^k \mathbf{r}_j^T \mathbf{A}_j^{-1} \mathbf{r}_j \quad (7)$$

where λ_k^W follows Chi-squared distribution with the DOF of MN . Additionally, the windowed innovation detector is represented as (Liu et al. 2019)

$$\eta_k^W = \left(\sum_{j=k-N+1}^k \mathbf{A}_j^{-1} \mathbf{r}_j \right)^T \left(\sum_{j=k-N+1}^k \mathbf{A}_j^{-1} \right)^{-1} \left(\sum_{j=k-N+1}^k \mathbf{A}_j^{-1} \mathbf{r}_j \right) \quad (8)$$

where η_k^W follows Chi-squared distribution with M degrees of freedom.

To quantitatively evaluate the accumulation performance of the two windowed detectors, the normalized spoofing energy of the snapshot detector is defined as the mean of the detection function λ_k

$$S_{\lambda_k} = E[\lambda_k] = E[\mathbf{r}_k^T \mathbf{A}_k^{-1} \mathbf{r}_k] = \boldsymbol{\mu}^T \mathbf{A}_k^{-1} \boldsymbol{\mu} \quad (9)$$

Since the DOF of λ_k is M , the standard deviation of it is $\sqrt{2M}$. Therefore, the noise energy in units consistent with that of S_{λ_k} , is the standard deviation of λ_k . Then, the INR can be defined as the ratio of the normalized spoofing energy to the noise energy

$$\text{INR}_{\lambda_k} = \frac{E[\lambda_k]}{\text{Std}\{\lambda_k\}} = \frac{\boldsymbol{\mu}^T \mathbf{A}_k^{-1} \boldsymbol{\mu}}{\sqrt{2M}} = \frac{S_{\lambda_k}}{\sqrt{2M}} \quad (10)$$

Under spoofing-present conditions, suppose that the innovation covariance matrix and offset caused by spoofing at every epoch within a window is constant, that is, $\mathbf{A}_{k-N+1} = \mathbf{A}_{k-N+2} = \dots = \mathbf{A}_k$ and $E(\mathbf{r}_{k-N+1}) = E(\mathbf{r}_{k-N+2}) = \dots = E(\mathbf{r}_k)$. In fact, for a slowly varying spoofing, the matrix and offset vary a little within a short time, so such an assumption is feasible. Based on the definition of the INR, the output INR of the two windowed detectors can be derived as

$$\begin{aligned} \text{INR}_{\lambda_k^W} &= \frac{E[\lambda_k^W]}{\text{Std}\{\lambda_k^W\}} = \frac{E\left[\sum_{j=k-N+1}^k \mathbf{r}_j^T \mathbf{A}_j^{-1} \mathbf{r}_j\right]}{\sqrt{2MN}} = \frac{\sum_{j=k-N+1}^k \boldsymbol{\mu}^T \mathbf{A}_j^{-1} \boldsymbol{\mu}}{\sqrt{2MN}} \\ &= \frac{N \boldsymbol{\mu}^T \mathbf{A}_k^{-1} \boldsymbol{\mu}}{\sqrt{2MN}} = \sqrt{N} \times \text{INR}_{\lambda_k} \end{aligned} \quad (11)$$

$$\begin{aligned} \text{INR}_{\eta_k^W} &= \frac{E[\eta_k^W]}{\text{Std}\{\eta_k^W\}} = \frac{E\left[\left(\sum_{j=k-N+1}^k \mathbf{A}_j^{-1} \mathbf{r}_j\right)^T \left(\sum_{j=k-N+1}^k \mathbf{A}_j^{-1}\right)^{-1} \left(\sum_{j=k-N+1}^k \mathbf{A}_j^{-1} \mathbf{r}_j\right)\right]}{\sqrt{2M}} \\ &= \frac{\left(\sum_{j=k-N+1}^k \mathbf{A}_j^{-1} \boldsymbol{\mu}\right)^T \left(\sum_{j=k-N+1}^k \mathbf{A}_j^{-1}\right)^{-1} \left(\sum_{j=k-N+1}^k \mathbf{A}_j^{-1} \boldsymbol{\mu}\right)}{\sqrt{2M}} = \frac{N \boldsymbol{\mu}^T \mathbf{A}_k^{-1} \boldsymbol{\mu}}{\sqrt{2M}} = N \times \text{INR}_{\lambda_k} \end{aligned} \quad (12)$$

From (10), (11), and (12), it can be seen that $\text{INR}_{\lambda_k^W}$ is $\sqrt{N}$ times more than INR_{λ_k} , while $\text{INR}_{\eta_k^W}$ is N times. While the INR should be used to compare decision statistics with the same underlying probability density functions (PDFs), it provides a second-order moment characterization of the two detectors. We can draw two conclusions based on comparing the INR values for the two spoofing detectors. First, a longer time window produces a higher INR, but the length of the time window is limited in practice. If the spoofing cannot be detected in time, it gradually induces the system outputs, and the innovation offset returns to zero again, causing the decrease in the INR and a failed detector. Thus, a suitable time window is required in the design of a spoofing detector. Second, the windowed innovation detector achieves a higher INR that is proportional to the square root of the time window length N , compared to the INR value for the windowed statistic detector.

Problem statement

As previously discussed, slowly varying spoofing attacks can gradually induce the coupled system to deviate from its true position over time by introducing small offsets in the innovation measurements, and even evade detection by the spoofing detector. Furthermore, this section presents a geometric analysis of the underlying reasons for the vulnerability of GNSS receivers to slowly varying spoofing attacks and provides a rigorous mathematical definition of slowly varying spoofing. By examining the effects of slowly varying spoofing on the innovation measurements and test statistics, we can gain deeper insights into the nature of this type of attack and develop more effective countermeasures to mitigate its impact.

Impact analysis of slowly varying spoofing

The definition of Mahalanobis distance for a given vector $\mathbf{x}$ is (Mahalanobis 1936)

$$D_M(\mathbf{x}) = \sqrt{(\mathbf{x} - E(\mathbf{x}))^T \Sigma^{-1} (\mathbf{x} - E(\mathbf{x}))} \quad (13)$$

where Σ^{-1} is the covariance matrix of $\mathbf{x}$. By comparing (5) and (13), it can be seen that the spoofing detection function λ_k is actually the squared Mahalanobis distance of the innovation vector γ_k . To visualize the effect of a slowly varying spoofing using a flat graph, it is assumed here that the length of γ_k is 2, that is, $\gamma_k = [\gamma_1, \gamma_2]^T$ and $A_k = [a_{11}, a_{12}; a_{21}, a_{22}]$. Here, γ_1 and γ_2 represent the pseudorange or pseudorange rate innovations from two distinct satellites. As each satellite's signal is processed through independent observation channels, γ_1 and γ_2 are mutually independent, with a covariance of 0, that is, $a_{12} = a_{21} = 0$. Under such an assumption, λ_k can be rewritten as

$$\lambda_k = r_1^2 + r_2^2 = (\gamma_1 / \sqrt{a_{11}})^2 + (\gamma_2 / \sqrt{a_{22}})^2 \quad (14)$$

where r_1 and r_2 are normalized innovations. The relationship among r_1 , r_2 , and the threshold T_D can be figured in a coordinate system, as shown in Fig. 2, where green denotes spoofing-absent conditions, while red denotes spoofing-present conditions, and the geometric centroid distance between them is μ . The solid points are the samples of (r_1, r_2) . Under spoofing-absent conditions, all green samples are located inside the green dashed-line circle, that is, $r_1^2 + r_2^2 < T_D$, indicating that the test statistic λ_k is below the threshold T_D . For spoofing-present conditions, the results depend on the location of the red samples relative to the detection threshold. In the left subfigure of Fig. 2, all of the red samples fall outside the green dashed-line circle, indicating that the test statistic λ_k exceeds the detection threshold T_D and the spoofing attack is detected with a detection probability of 100%.

In the middle subfigure, only some of the spoofing can be detected. As the geometric centroid of the red points still lies outside the threshold, the windowed detector can take effect by accumulating time and suppressing noise. In the right subfigure, the spoofed central point is encircled by the threshold, and suppressing noise is detrimental to detection. In this case, the windowed detector cannot work effectively.

Definition of the slowly varying spoofing

The above analysis of the three spoofing cases shows that whether a spoofing can be detected is determined by the Mahalanobis distance of the innovation offset caused by the spoofing. From Fig. 2, it can be seen that for the three spoofing cases, the relationships between the detection statistic and threshold are

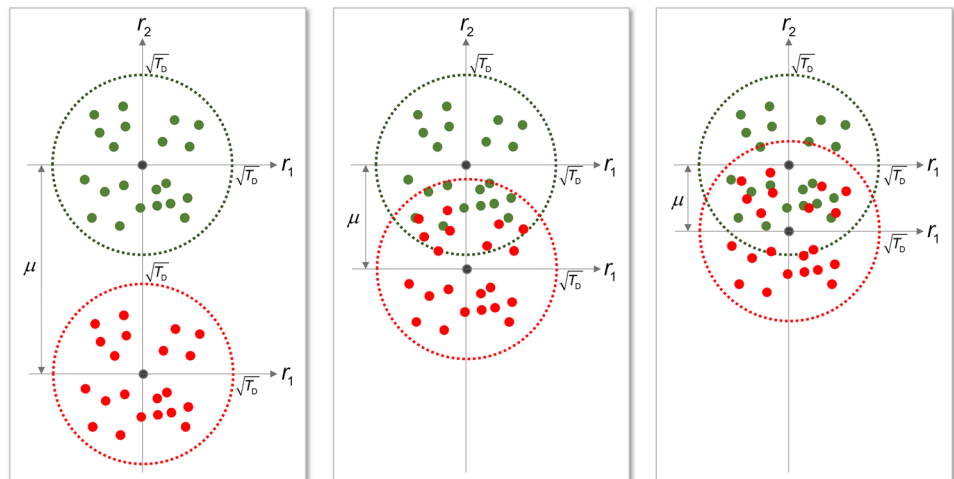
$$\begin{cases} 2\sqrt{T_D} < \sqrt{\lambda_k} < \infty, & \text{high - drift spoofing} \\ \sqrt{T_D} < \sqrt{\lambda_k} \leq 2\sqrt{T_D}, & \text{mid - drift spoofing} \\ 0 < \sqrt{\lambda_k} \leq \sqrt{T_D}, & \text{low - drift spoofing} \end{cases} \quad (15)$$

The three inequalities in (15) provide quantitative definitions of suddenly and slowly varying spoofing based on the geometric interpretation of the spoofing detection metric. A sudden varying spoofing corresponds to a high-drift spoofing, while a slowly varying spoofing corresponds to a mid-drift or low-drift spoofing.

Proposed weighted Kalman gain detector

Based on the analysis presented in the problem statement section, we have identified a significant challenge in detecting low-drift spoofing attacks using conventional windowed spoofing detectors. We propose a novel weighted Kalman

Fig. 2 Three spoofing cases. Each subplot corresponds to $\mu > 2\sqrt{T_D}$ (left), $\sqrt{T_D} < \mu \leq 2\sqrt{T_D}$ (middle), and $0 < \mu \leq \sqrt{T_D}$ (right), with the innovation samples marked in green (spoofing-absent) or red (spoofing-present)



gain detector to achieve a higher detection sensitivity than conventional windowed detectors to address this challenge.

In the computation of the innovation $\mathbf{y}_k = \mathbf{Z}_k - \mathbf{H}_k \hat{\mathbf{X}}_{k/k-1}$, the measurement vector $\mathbf{Z}_k$ consist of pseudorange and pseudorange rate observations at epoch k , arranged in the order of satellite observations. Therefore, $\mathbf{Z}_k$ can be partitioned based on the satellite sequence as $\mathbf{Z}_k = [\mathbf{Z}_k^{(1)} \mathbf{Z}_k^{(2)} \dots \mathbf{Z}_k^{(L)}]^T$, where L represents the number of visible satellites. Correspondingly, the innovation vector $\mathbf{y}_k$ can also be expressed in a partitioned form based on the satellite index as $\mathbf{y}_k = [\mathbf{y}_k^{(1)} \mathbf{y}_k^{(2)} \dots \mathbf{y}_k^{(L)}]^T$. The measurement matrix $\mathbf{H}_k$ is a block matrix that corresponds to the satellite indices, and the detailed proof for this can be found in Noureldin et al. (2013). To facilitate the analysis of each satellite, a sequential filtering framework is adopted, where the superscript (i) in the following equations denotes the sequential number of the observing satellite. Consequently, the innovation is rewritten as

$$\mathbf{y}_k^{(i)} = \mathbf{Z}_k^{(i)} - \mathbf{H}_k^{(i)} \hat{\mathbf{X}}_k^{(i-1)}, \quad i = 1, 2, \dots, L \quad (16)$$

where $\mathbf{Z}_k^{(i)}$ is the pseudorange and pseudorange rate information corresponding to the i -th satellite. $\mathbf{H}_k^{(i)}$ is the measurement matrix corresponding to the i -th satellite. $\mathbf{y}_k^{(i)}$ is the innovation corresponding to $\mathbf{Z}_k^{(i)}$. $\hat{\mathbf{X}}_k^{(i-1)}$ is the state vector updated by the observations from the $(i-1)$ -th satellite. Specifically, $\hat{\mathbf{X}}_k^{(0)} = \hat{\mathbf{X}}_{k/k-1}$ represents the prior estimate of the state vector at epoch k , which is the predicted state vector obtained from the INS outputs. Consequently, the covariance matrix of $\mathbf{y}_k^{(i)}$ is $\mathbf{A}_k^{(i)}$, from which the test statistic for the i -th satellite can be obtained

$$\eta_k^{(i)W} = \left(\sum_{j=k-N+1}^k (\mathbf{A}_j^{(i)})^{-1} \mathbf{y}_j \right)^T \left(\sum_{j=k-N+1}^k (\mathbf{A}_j^{(i)})^{-1} \right)^{-1} \left(\sum_{j=k-N+1}^k (\mathbf{A}_j^{(i)})^{-1} \mathbf{y}_j \right) \quad (17)$$

where the DOF of Chi-square variable $\eta_k^{(i)W}$ is 2. The total number of satellites is L , and the allowed FAR is P_F . The FAR is assigned to each satellite equally, thus the relationship between the FAR and the detection threshold $T^{(i)}$ for a single satellite is

$$\frac{P_F}{L} = P\left\{ \eta_k^{(i)W} > T^{(i)} | H_0 \right\} = \int_{T^{(i)}}^{\infty} \chi^2(\eta, 2) d\eta = \int_{T^{(i)}}^{\infty} \frac{1}{2} e^{-\frac{\eta}{2}} d\eta = e^{-\frac{T^{(i)}}{2}} \quad (18)$$

Solving the above equation, we obtain the detection threshold for the innovation of a single satellite as

$$T^{(i)} = 2 \ln \left(\frac{L}{P_F} \right) \quad (19)$$

When a slowly varying spoofing is present, the innovation offset is typically very small, making it difficult for

the windowed innovation detector to exceed the detection threshold. However, the distance between the test statistic and the detection threshold reflects the confidence level of the innovation measurement, with a smaller distance indicating a higher likelihood of spoofing. Thus, the distance can be utilized to weight the innovation measurement. The weight factor can be determined empirically as

$$\alpha^{(i)} = \begin{cases} 1 - \sqrt{\frac{\eta_k^{(i)W}}{T^{(i)}}}, & \eta_k^{(i)W} \leq T^{(i)} \\ 0, & \eta_k^{(i)W} > T^{(i)} \end{cases} \quad (20)$$

The above equation indicates the likelihood of a spoofing attack at the level of the different measurements. The weight factor is zero if a satellite-level decision statistic is above the threshold. It increases as the trust in the satellite increases. As such, the weight factor reflects the level of trust in the received satellite observations and can be interpreted as a trust factor. By incorporating the weight factor into the measurement update loop of the Kalman filter, we obtain a measurement update equation that is capable of detecting slowly varying spoofing attacks more effectively

$$\begin{cases} \mathbf{K}_k^{(i)} = \mathbf{P}_k^{(i-1)} (\mathbf{H}_k^{(i)})^T \left[\mathbf{H}_k^{(i)} \mathbf{P}_k^{(i-1)} (\mathbf{H}_k^{(i)})^T + \mathbf{R}_k^{(i)} \right]^{-1} \\ \hat{\mathbf{X}}_k^{(i)} = \hat{\mathbf{X}}_k^{(i-1)} + \alpha^{(i)} \mathbf{K}_k^{(i)} \mathbf{y}_k^{(i)} \\ \mathbf{P}_k^{(i)} = (\mathbf{I} - \alpha^{(i)} \mathbf{K}_k^{(i)} \mathbf{H}_k^{(i)}) \mathbf{P}_k^{(i-1)} \end{cases} \quad (21)$$

Based on (20) and (21), it can be deduced that in the presence of slowly varying spoofing, $\eta_k^{(i)W}$ approaches $T^{(i)}$, and $\alpha^{(i)}$ tends to zero. This diminishes the impact of the corresponding satellite observations on the estimation of the Kalman filter, thereby mitigating the degree of deviation caused by spoofing on the estimation results. Consequently, this leads to a prolonged period of accurate estimation results in the presence of spoofing, facilitating the accumulation of spoofing offset within the innovation. Eventually, this accumulation causes the test statistic to exceed the detection threshold. This concept is illustrated in Fig. 3, where the blue bidirectional arrow represents the range of measurement errors. As shown in the figure, when a spoofing attack occurs at epoch k , the innovation offset is initially small and falls below the measurement noise, rendering it undetectable. This causes the position estimate of the GNSS to gradually deviate, leading to the output of the integrated filter gradually departing from the true position and ultimately coinciding completely with the spoofing trajectory. However, the trajectory indicated by the green triangle shows that the inclusion of the

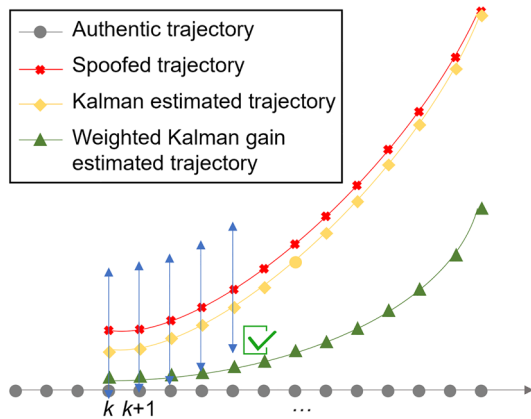


Fig. 3 Schematic of the effect of spoofing on the filtered output

trust factor mitigates the impact of the spoofing on the filter output in the short term. Over time, the innovation offset gradually exceeds the measurement noise and can be detected.

At epoch k , after sequential filtering has traversed all visible satellites, the final test statistic is obtained

$$\eta_k^{LW} = \sum_{i=1}^L \eta_k^{(i)W} \quad (22)$$

where the DOF of Chi-square variable η_k^{LW} is $2L$. The corresponding detection threshold T_d can be determined as

$$P_F = \int_{T_d}^{\infty} \chi^2(\eta, 2L) d\eta \quad (23)$$

where η is the samples of η_k^{LW} . Based on the above analysis, the process for detecting slowly varying spoofing attacks with the inclusion of a trust factor can be summarized as indicated in Fig. 4.

Simulation performance

In this section, 1000 Monte Carlo simulations were carried out to compare the detection performances of the above three spoofing detectors: windowed statistic detector, windowed innovation detector, and weighted gain detector.

Simulation conditions

The parameters of the simulated INS and GNSS systems are presented in Table 1. In the simulations, the TCS undergoes a 200-s motion, starting at 120° E and 30° N, and then horizontally moving at a speed of 100 m/s with a yaw angle of 40° .

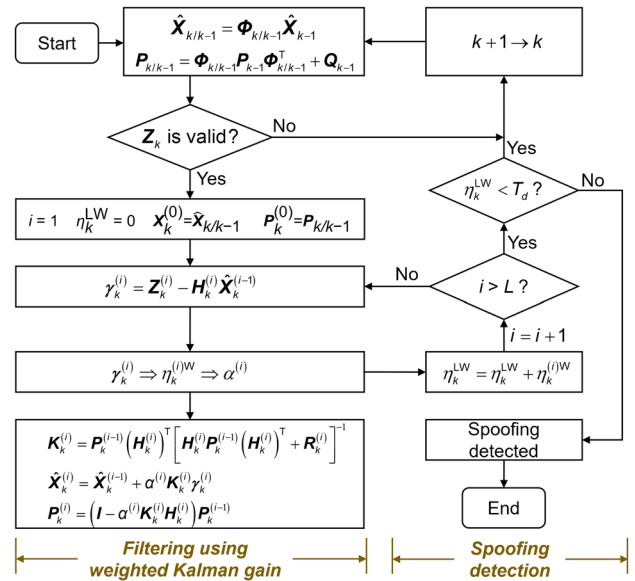


Fig. 4 Flowchart of the proposed weighted Kalman gain detector. The left side shows the filtered output obtained from the weighted Kalman gain, and the right side shows the computation of the test statistic and the implementation of the spoofing detection

Next, based on the simulated attitude, velocity, and position (AVP) data and the INS error parameters specified in Table 1, the measurements of angular rate and acceleration for the INS are simulated in a reverse manner. Similarly, the GNSS measurements of pseudorange and pseudorange rate are generated using a similar reverse approach. To simulate the effect of spoofing, during the latter 100 s of the simulation, a latitude component velocity offset is added to the original simulated AVPs to represent the spoofing velocity offset. Consequently, the position information is also modified accordingly as the velocity offset is introduced. The AVPs with the velocity offset are used to generate the simulated GNSS observations for the latter 100 s. This setup effectively simulates a case where the GNSS receiver is influenced by spoofing signals starting

Table 1 Sensors parameters

Sensor items	Parameters	Values
INS	Gyro bias	$0.5^\circ/\text{h}$
	Gyro random walk	$0.15^\circ/\text{sqrt}(\text{h})$
	Accelerometer bias	0.05 mg
	Accelerometer random walk	$0.06 \text{ m/s}^2/\text{sqrt}(\text{h})$
	Sampling rate	100 Hz
GNSS	Pseudorange error (3σ)	5 m
	Pseudorange rate error (3σ)	0.2 m/s
	Constellation	Week 949 almanac
	Number of observed satellites	10
	Measurement update rate	10 Hz

from the 100th second, resulting in a deviation between the computed and true velocities. According to the definition in (15), it can be computed that a high-drift spoofing holds the velocity offset δv higher than 0.86 m/s, a mid-drift spoofing holds the velocity offset between 0.43 and 0.86 m/s, and a low-drift spoofing holds the velocity offset lower than 0.43 m/s. To cover these three spoofing items, ten different spoofing velocity offsets are simulated, ranging from 1 to 0.1 m/s at 0.1 m/s intervals. Additionally, the length of the time window N is set to 10 (Zhang et al. 2022), and the FAR P_F is 10^{-5} (Wang et al. 2022).

Detection probability

Figure 5 shows the normalized innovations generated by the three detectors for the spoofing with a minimum drift of 0.1 m/s. The three subfigures from left to right correspond to the windowed statistic detector, windowed innovation detector, and the weighted gain detector proposed in this study. It can be observed that the normalized innovations generated by the window statistic detector deviate only slightly from the innovations obtained under no spoofing, with a maximum value of 3.9, indicating minimal offset caused by spoofing intrusion. On the other hand, the windowed innovation detector exhibits some degree of cumulative effect on the innovations, with the maximum value accumulating to 4.2, indicating a certain level of deviation from the innovations obtained under no spoofing. The weighted gain detector proposed in this study mitigates the effect of spoofing on the filtered output through a weighted factor, which further accumulates the deviation of the innovations and increases it to 4.9. The larger the absolute value of the deviation of the innovations, the larger the value of the constructed detection statistics, and the more likely it is to exceed the detection threshold, resulting in successful detection. Therefore, based on the deviation of the innovations, the windowed

innovation detector exhibits better detection performance than the windowed statistic detector, which is consistent with the INR analysis results. The weighted gain detector proposed in this study generates the largest deviation of the innovations and exhibits the best detection performance.

To demonstrate the detection capabilities of the three detectors more clearly, Fig. 6 shows the relationship between the detection statistics generated by the three detectors and the detection thresholds under high-drift spoofing with $\delta v = 1$ m/s, mid-drift spoofing with $\delta v = 0.5$ m/s, and low-drift spoofing with $\delta v = 0.1$ m/s. It can be seen that for high-drift spoofing, all three detectors can achieve successful detection. However, the degree to which the window statistic detector exceeds the detection threshold is very small, almost at the detection threshold. For mid-drift spoofing, the windowed statistic detector completely fails to detect the spoofing. As for low-drift spoofing, the windowed innovation detector also fails to detect it, while the weighted gain detector proposed in this study is still able to successfully detect the occurrence of the spoofing.

To quantitatively evaluate the detection performance of a given detector, the detection probabilities of the three detectors were calculated for different spoofing drifts, as shown in Fig. 7. The detection probability is defined as the ratio of the number of times spoofing is detected to the total number of simulations. The larger this value, the better the detection performance. It can be seen that the windowed statistic detector only exhibits high detection probability for high-drift spoofing, while both the windowed innovation detector and the weighted gain detector proposed in this study can effectively detect all three types of spoofing. However, for any spoofing velocity drift, the weighted gain detector proposed in this study has the highest detection probability. Especially for the extremely low-drift spoofing of $\delta v = 0.1$ m/s, the detection probability of the windowed innovation detector is 59.9%, while

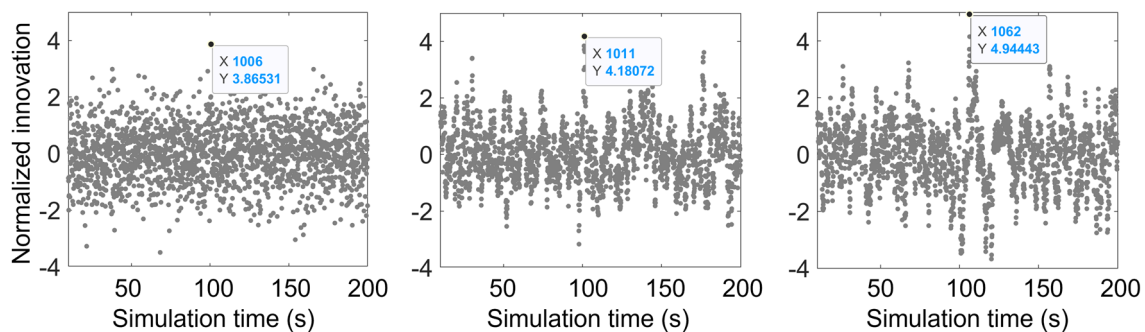


Fig. 5 Normalized innovations generated by the three detectors with $N=10$ and $P_F=10^{-5}$. Each subplot corresponds to the windowed statistic detector (left), windowed innovation detector (middle), and weighted gain detector (right)

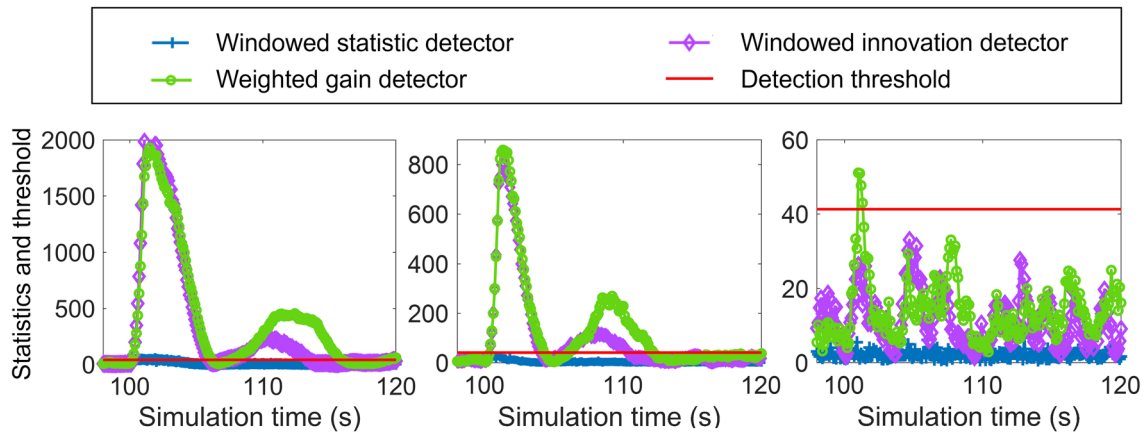


Fig. 6 Statistics and threshold of the three detectors with $N=10$ and $P_F=10^{-5}$. Each subplot corresponds to $\delta v=1$ m/s (left), $\delta v=0.5$ m/s (middle), and $\delta v=0.1$ m/s (right)

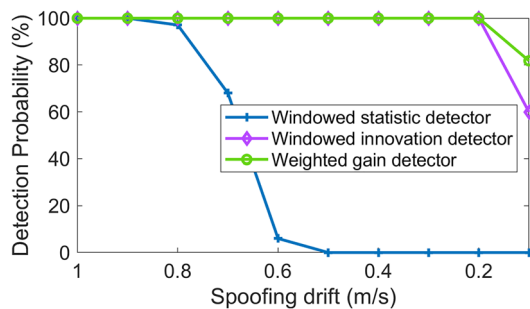


Fig. 7 Detection probabilities versus spoofing drifts of the three detectors with $N=10$ and $P_F=10^{-5}$

the detection probability of the proposed weighted gain detector is 81.8%. Therefore, our proposed weighted gain detector has a significant advantage in detecting low-drift spoofing.

Receiver operating characteristic

To further verify detection probabilities of the three detectors under different false alarm rate settings, we plotted the receiver operating characteristic (ROC) curves for the three detectors, as shown in Fig. 8. It can be observed that our proposed weighted gain detector consistently outperforms the other two detectors under different spoofing drifts. In particular, for low-drift spoofing, it exhibits the fastest rising trend in the ROC curve.

Experiment verification

To further demonstrate the performance of the proposed detector in practical environments, we conducted an experiment in Nankai University. More detailed descriptions of the experiment and results are listed below.

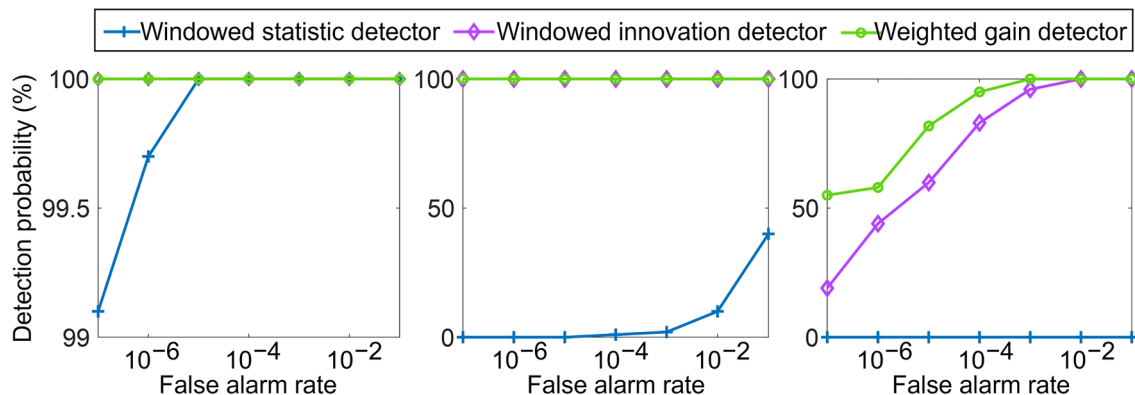


Fig. 8 ROC curves of the three detectors for different spoofing drifts. Each subplot corresponds to $\delta v=1$ m/s (left), $\delta v=0.5$ m/s (middle), and $\delta v=0.1$ m/s (right)

Experimental settings

The experimental setup, depicted in the left subplot of Fig. 9, consists of a GNSS receiver (NEO-M8T), an INS (STIM-300), and a processing computer. The GNSS and INS parameters are the same as those used for the simulation, as presented in Table 1. The length of time window and FAR are the same as in the simulations. During experiment, the GNSS receiver acquires the ephemeris, pseudoranges, and pseudorange rates from Global Positioning System (GPS) L1 signals, and provides a pulse per second (PPS) signal that serves as the external trigger signal for the INS, ensuring time synchronization between them. The INS generates acceleration and angular velocity data. Finally, these data are processed by the tightly coupled filtering and spoofing detection algorithms on the processing computer.

In the right subplot of Fig. 9, the moving trajectories during the experiment are depicted, with the blue line representing the authentic trajectory and the red line representing the spoofed trajectory. The total experiment duration is 200 s, and around 120 s, the GNSS antenna is gradually moved along the trajectory indicated by the red line in the figure to simulate the spoofed trajectory.

Experimental results

Figure 10 illustrates the test statistics and thresholds of the three detectors. The windowed statistic detector exhibits a low frequency and degree of test statistics exceeding the thresholds, making it difficult to detect spoofing. Moreover, the detector shows a significant lag. In contrast, the windowed innovation detector promptly detects spoofing and effectively accumulates the innovation offset, resulting in larger test statistics. Our proposed weighted gain detector weakens the harmful innovation's influence on the filtering result through weighted factors, further accumulates the innovation offset, and generates larger test statistics than the windowed innovation detector. This enhances the detection sensitivity and is advantageous for detecting lower drift spoofing.

To quantitatively compare the detection performance of the three detectors, we adopt three evaluation metrics: detection ratio, time to alarm, and alarm duration. The detection ratio r_d is defined as the maximum ratio of test statistics exceeding the threshold

$$r_d = \max \frac{\lambda_k - T_D}{T_D} \quad k = 1, 2, \dots, K \quad (24)$$

Fig. 9 Experimental settings. Each subplot corresponds to the experiment equipment (left) and the moving trajectories (right)

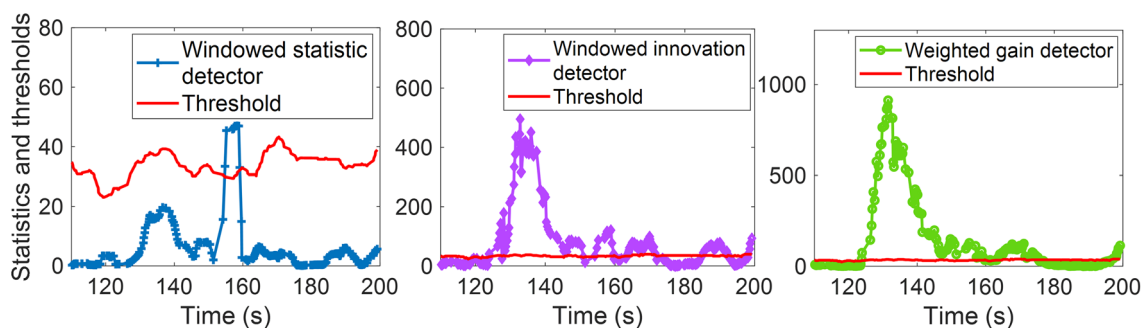
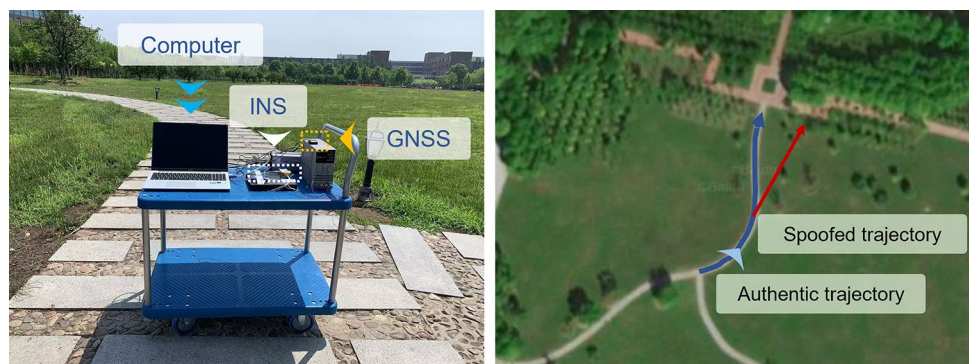


Fig. 10 Statistics and thresholds with $N=10$ and $P_F=10^{-5}$. Each subplot corresponds to windowed statistic detector (left), windowed innovation detector (middle), and weighted gain detector (right)

where K is the total number of measurement update points. A higher r_d indicates a greater degree of test statistics exceeding the threshold, which corresponds to a higher accumulation of innovation offset and better detection sensitivity, implying better detection performance of the detector. The time to alarm is defined as the duration between the occurrence of spoofing and the detection of spoofing by the detector, reflecting the timeliness of the detector in detecting spoofing. A shorter time to alarm can help users to promptly identify spoofing and take measures to mitigate the adverse effects of spoofing. The alarm duration is defined as the duration during which the test statistics exceed the threshold after the occurrence of spoofing, and a longer alarm duration indicates a more reliable warning indication. The statistical results of these three evaluation metrics for the different detectors are presented in Table 2. Our proposed weighted gain detector exhibits the highest detection ratio, shortest time to alarm, and longest alarm duration, indicating superior detection performance compared to the other two detectors.

Conclusions and discussions

Detecting slowly varying spoofing presents the most significant challenge in GNSS spoofing detection. Through a comparison of the output INR, we determined that the windowed innovation detector exhibits higher sensitivity than the windowed statistic detector. Additionally, we established a quantitative definition for slowly varying spoofing. Building on these findings, we proposed an improved detector for slowly varying spoofing in GNSS/INS tightly coupled navigation systems. This detector leverages a weighted Kalman gain to enhance detection sensitivity. Our proposed detector addresses the limitations of conventional EKF-based detectors by incorporating the impact of spoofing on innovation offsets and applying weight to the EKF gain based on the normalized distance between the test statistic of each satellite and the detection threshold. Simulation results indicate that our proposed weighted gain detector significantly improves the detection probability for extremely low-drift spoofing, increasing it from 59.9% with the windowed innovation detector to 81.8%. Furthermore, experimental results

validate the advantages of our proposed detector in practical environments. This novel approach to detecting slowly varying spoofing significantly contributes to the field of GNSS security. Future work will focus on enhancing the proposed detector's robustness against different types of spoofing attacks and evaluating its performance in more complex environments.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10291-023-01594-3>.

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Authors contributions Conceptualization was done by XJ; methodology was done by XJ and SL; simulations and experiments were carried out by XJ; writing—original draft preparation was done by XJ; writing—review and editing was done by XZ; funding acquisition was done by SZ; supervision was done by XZ.

Data availability All data generated or analyzed during this study are included in the supplementary information files.

Declarations

Competing interests The authors declare no competing interests.

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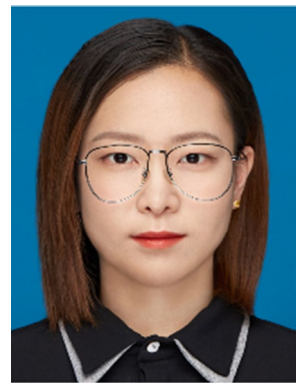
Table 2 Experiment results

Detector item	Detection ratio	Time to alarm (s)	Alarm duration (s)
Windowed statistic	0.6	24.8	4.9
Windowed innovation	13.0	3.9	47.6
Weighted gain	23.5	3.8	51.4

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Xiaoqin Jin received a BS degree from the College of Information and Communication Engineering, Harbin Engineering University, Harbin, China, in 2017, and an MS degree from the School of Electronic and Information Engineering, Beihang University, Beijing, in 2020. She is pursuing a PhD degree with the College of Artificial Intelligence, Nankai University. Her research interest is GNSS signal processing.



Xiaoyu Zhang is a full professor in the Institute of Robotics and Automatic Information System, College of Artificial Intelligence, Nankai University, Tianjin, China. His research interests include navigation, guidance, control, and artificial intelligence.



Shuaiyong Zheng received the PhD degree from Beihang University in 2021. He currently works as a Communication Engineering lecturer with the School of Integrated Circuit Science and Engineering, Tianjin University of Technology. His research interests cover satellite-based augmentation, inertial, and integrated navigation systems.



Shoupeng Li received his PhD degree in aeronautical and astronautical science and technology from Harbin Institute of Technology, in 2022. He is a postdoctoral researcher at the Institute of Modern Optics, Nankai University. His main research interests are related to nonlinear filtering theory, target tracking, integrated navigation, inertial navigation, and initial alignment of strap-down inertial navigation system.